

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

MSE 2201
Phase Diagrams and Transformations

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What is polymorphism? "Pure iron can show both magnetic and non-magnetic behaviour depending on the temperature"—Explain this statement. (15)
- (b) What is Gibbs' phase rule? Determine the degree of freedom (DF) of the following systems: (12)
- (i) Pure frozen solid water
 - (ii) A mixture of liquid and vapour water
 - (iii) A mixture of vapour liquid of water and Hexane (C_6H_{14})
- (c) Determine the degree of freedom in a Cu-60% Ni alloy at 1000 °C, 1320 °C and 1400 °C respectively. Assuming that, the Cu and Ni are completely soluble in solid and liquid state. Given, the melting point of Cu is 1085 °C and Ni is 1455 °C. (8)
- Q 2. (a) What is cored structure? Explain with necessary phase diagram how non-equilibrium cooling happens. What is the significant difference found in microstructure at different temperatures due to non-equilibrium cooling? (15)
- (b) Write short notes on: (10)
- (i) Solid solution
 - (ii) Invariant point
- (c) Why is peritectic reaction called incomplete reaction? How can magnetic properties be found by peritectic reaction? (10)
- Q 3. (a) Derive a mathematical way to find out the amount of pure phases present in a certain alloy at certain variables. (10)
- (b) Consider an alloy A-B in which the melting points of A and B are 400 °C and 900 °C respectively. Calculate the amount of phases and chemical compositions of A and B at 500 °C. Draw necessary diagram to solve the problem. (15)
- (c) Draw and explain the cooling curve of hypoeutectic and hypereutectic alloy. (10)
- Q 4. (a) What is Hume-Rothery rule? How can an intermediate compound be detected by this rule? Do you think Cu_9Al_4 , Cu_3Si , Cu_3Ga , Cu_3Sc form intermediate compounds? (15)
- (b) What is martensite? Classify the types of martensite. Explain with necessary microstructures, which type of martensite is most preferable and why? (10)
- (c) Draw a typical TTT diagram. How pearlite can be formed from bainite—explain with the help of TTT diagram. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Derive an equation for calculating the critical nucleus size during solidification of a pure liquid metal. How does the degree of supercooling affect the critical size of the nucleus? (18)
- (b) Differentiate between homogeneous nucleation and heterogeneous nucleation for the solidification of a pure metal. (7)
- (c) Describe the spiral growth of crystal with proper illustration. (10)
- Q 6. (a) Derive Fick's first law for the non-steady state diffusion. (15)
- (b) Discuss the factors that influence on diffusion process. (12)
- (c) Describe an experimental investigation of self-diffusion in a pure metal. (8)
- Q 7. (a) What is Gibbs' triangle? Show that, if any two ternary alloys are mixed together, the composition of the mixture will lie on a straight line joining the original two components. (12)
- (b) Explain the parallel line method for the calculation of ternary alloy concentrations. (8)
- (c) Discuss the advantages and limitations of (i) isothermal and (ii) vertical sections of a ternary phase diagram. (15)
- Q 8. (a) With a neat sketch, show that when two phases coexist in a condensed ternary system, there can only be two degree of freedom. (10)
- (b) Explain the isothermal equilibrium solidification of a ternary solid solution alloy with neat sketches. (20)
- (c) Draw a hypothetical concentration triangle of a ternary system showing complete solid insolubility. (5)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

MSE 2203
Mechanical Behavior of Materials

Time: 3 hours

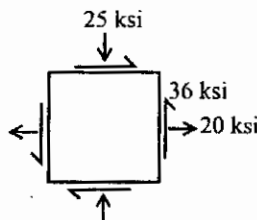
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Define 'strength coefficient' and 'strain hardening exponent' of material. (08)
(b) Show that necking begins when the true strain is equal to the strain hardening exponent of material. (12)
(c) Two points on a stress-strain curve for a material are $\sigma = 278$ MPa at $\epsilon = 0.08$ and $\sigma = 322$ MPa at $\epsilon = 0.16$. Find the strain hardening exponent and strength coefficient of the material and predict σ at $\epsilon = 0.20$. [Assume Power Law for calculation] (15)
- Q 2. (a) Define hydrostatic and deviatoric state of stress. (08)
(b) How do you predict failure criteria of ductile material using Von Mises' yield criteria. (12)
(c) A structure element is subjected to the state of stress shown below: (15)



Materials specification: BSI G60 400W, Tensile strength 78,000 psi, yield strength 60,000psi, Elongation 15% in 2 inch. Determine the factor of safety with respect to failure by yielding according to:

- (i) Maximum shear stress theory
(ii) Maximum distortion energy (Von Mises') theory

- Q 3. (a) Write down the basic differences between edge and screw dislocation. (08)
(b) "Density of dislocation increases during plastic deformation"—explain it with the concept of Frank-Read source. (12)
(c) What is meant by "critical resolved shear stress"? Determine the tensile stress that is applied along the $[1\bar{1}0]$ axis of a silver crystal to cause slip on the $(1\bar{1}\bar{1})[0\bar{1}1]$ system. The critical resolved shear stress is 6 MPa. (15)
- Q 4. (a) What is "strain hardening" of material? Draw and explain the shear stress (τ) vs shear strain(γ) curve when a single crystal material undergoes strain hardening. (13)
(b) Define "recrystallization temperature". What will happen if a strain hardened material undergoes annealing? (12)
(c) Write down the effects of "dislocation pile-up" in the crystal. (10)

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) State and explain the Griffith Theory and derive an equation for stress required to propagate a crack at plain strain conditions in a brittle material as a function of size of the microcrack. (15)
- (b) Write short note on "Fractography". (10)
- (c) Etching removes surface defects and relieves surface residual stress. A 5 mm thick borosilicate glass after etching has cracks inside ranging from 2 to 5 μm in length. Calculate the surface energy of glass using following parameters: Fracture strength is 90 MPa, Young's Modulus is 60 GPa and Poisson's ratio is 0.3. (10)
- Q 6. (a) Derive an equation for strain energy release rate and hence develop a relationship between critical strain energy release rate and critical load at which the crack runs unstably to fracture. (18)
- (b) Demonstrate the crack-tip-opening displacement (CTOD) concept and crack opening displacement (COD) model with necessary figures. (17)
- Q 7. (a) What is creep? Explain the creep process with the aid of a creep curve. (15)
- (b) Explain the stress-rupture test of creep. (10)
- (c) Write short notes on: "High temperature alloys". (10)
- Q 8. (a) What is Fatigue? Explain the basic necessary factors to cause fatigue failure. (10)
- (b) Explain the Goodman diagram of dependency of limiting range of stress on mean stress. (15)
- (c) A 4340 steel bar is subjected to a fluctuating axial load that varies from a maximum of 330 kN tension to a minimum of 110 kN compression. The mechanical properties of the steel are: $\sigma_u = 1090$ MPa; $\sigma_o = 1010$ MPa; $\sigma_e = 510$ MPa (symbols have their usual meaning). Determine the bar diameter to give infinite fatigue life based on a safety factor of 2.50. Assuming the bar has a constant cylindrical cross section of A. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

CSE 2227
Programing Techniques

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Classify data types in C. (05)
- (b) Identify and correct the errors for the following statements: (20)
- (i) scanf ("d", value);
 - (ii) printf (" The product of %d and %d" \n, x, y);
 - (iii) first number + second number = sum of numbers;
 - (iv) if (number = > largest) largest = = number;
 - (v) /* program to determine the largest */;
 - (vi) scanf ("%d", an Integer);
 - (vii) printf ("Remainder of %d divided by %d is \n", x, y, x%y);
 - (viii) if (x=y)
 - (ix) printf (%d is equal to %d \n", x, y);
 - (x) printf ("The sum is %d \n," x + y);
 - (xi) printf ("The value you entered is: %d \n, & value);
- (c) Write a program in C that can solve the equation $ax^2 + bx + c = 0$. (10)
- Q 2. (a) What do you mean by logical values? (07)
- (b) Define rational operator and explain it with example(s). (07)
- (c) Write a program that will obtain the length and width of a rectangle from the user and compute its area and perimeter. (06)
- (d) Describe if structure with an example. (05)
- (e) Write a program in C that can calculate the factorial of N, where N is an integer number. (10)
- Q 3. (a) Write the difference between (i) one-dimensional and (ii) two-dimensional array. (08)
- (b) Suppose you have 60 (sixty) students in your class and all of you have given an examination on two subjects whose titles are (i) CSE 2227 and (ii) Math 2207. Now, write a C program that can store 'Roll', 'Subject title' and 'Marks' for the exam. (15)
- (c) Sort the following symbols according to precedence and associativity: (07)
- $=$, $+$, $+$, $++$ (postfix), $--$ (prefix), $<$, $\times$.
- (d) Write a program that demonstrates the difference between predecrementing and postdecrementing using the decrement operator. (05)

- Q 4. (a) What is a function? Explain. (06)
- (b) Write the difference between (i) standard library function and (ii) user-defined function. (08)
- (c) Write a C program for following problems: (21)
- (i) Search a specific number of an array element and say its position if it is found
 - (ii) Calculate the sum of border element of two dimensional array
 - (iii) Add two matrices

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) What will the following repetition structures print? (15)
- (i) for ($i = x; i \leq 4 * x * y; i += \frac{y}{x}$), here $x = 2, y = 10$.
 - (ii) for ($i = 1; i \leq 100; i++$)
 - (iii) for ($i = 7; i \leq 77; i += 7$)
 - (iv) for ($i = 44; i \geq 0; i -= 11$)
 - (v) for ($i = 2; i \leq 80; i += 5$)
- (b) A person invests \$1000.00 in a savings account yielding 5% interest. Assuming that all interest is left on deposit in the account, calculate and print the amount of money in the account at the end of each year for 10 years. (08)
- (c) Write a program that calculates and prints the sum of the even integers from 0 to 20. Write a program that prints the product of odd integers from 1 to 15. (12)
- Q 6. (a) Explain about the following built-in string functions of C (12)
- (i) strcpy () (ii) strcat () (iii) strlen () (iv) strcmp ()
- (b) Write a program to count the number of characters, words, lines and sentences of a text file. (10)
- (c) Write a program in C that can find a word from a string. (13)
- Q 7. (a) Write the difference between (i) array of structure and (ii) linked-list. (10)
- (b) What do you mean by dynamic memory allocation? Explain. (07)
- (c) Write a program to read a string or line and print it in the reverse order. (10)
- (d) Write a program to convert a string into upper case letter. (08)
- Q 8. (a) Define pointer. How will you implement pointer in programming? Write a program that can create a list with pointer and insert a data into it and delete a data from it. (15)
- (b) Write a program to scan a string data and print it using pointer. (10)
- (c) Using binary search method, search a number from linear array. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

Math 2227
Statistics and Numerical Analysis

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Normal distribution table (z-table), t-table, χ^2 -table are provided in the exam hall

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Define interpolation with example. Draw the forward diagonal difference table for six arguments. (09)
(b) Deduce Newton's general interpolation formula for divided difference. (13)
(c) Applying Lagrange's interpolation formula to find the form of the function $f(x)$ using the following table: (13)

x	-1	0	2	5
$f(x)$	-7	1	11	101

- Q 2. (a) What do you mean by root of a nonlinear equation; explain with figure. (06)
(b) Derive False Position (regula falsi) formula with graphical illustration. (11)
(c) Find the root of the equation by Newton-Raphson method correct to two decimal point. (15)
 $f(x) = x^2 + 4 \sin x$
(d) State 1 (one) limitation of: (i) False position (ii) Newton-Raphson method. (03)

- Q 3. (a) Let $f(x) = \log x$. Find the values of $f(x)$ for $x = 500 + n \cdot 10$; $n = 0, 1, 2, 3, 4, 5$. Using these data find $f'(500)$ and $f''(500)$ and correct up to six decimal places. Also find the percentage of relative error. (20)
(b) The population of a town in the decennial census was as given below. Estimate the population for the year 1985. (15)

Year	1970	1980	1990	2000	2010
Population (in million)	70	90	120	140	160

- Q 4. (a) What is the necessity of numerical integration? (5)
(b) Evaluate $\int_0^{\pi/2} e^{\sin x} dx$ by using Simpson's $3/8$ rule. (13)
(c) Using Gauss elimination method, solve the following system of linear equations: (17)

$$\begin{aligned} x + y + z &= 6 \\ 2x - y + 3z &= 4 \\ 4x + 5y - 10z &= 13 \end{aligned}$$

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Find the first four central moments. Hence find standard deviation, coefficient of variation and coefficient of Kurtosis. Moreover, draw the Histogram of cumulative frequency distribution. (18)

Class Mark	5	10	15	20
Observation	2	4	3	1

- (b) Test whether the following $P(x)$ be a probability mass function or not. If possible, find first four moments of the following function. Hence find variance and the coefficient of skewness of the function. (17)

x	0	1	2	3
$P(x)$	0.1	0.2	0.4	0.3

- Q 6. (a) Test whether the following functions are Probability distribution function or not. (11)

(i)

x	-1	0	1	2
$P(x)$	-3k	2k	4k	0.5

(ii)

x	-2	0	2	4
$P(x)$	2k	2k	k	0.5

Hence if possible, find $E(2x + 3)$ and $P(-2 \leq x < 2)$ for each instance.

- (b) Find the mean and variance of Binomial distribution. (12)
- (c) Show that the coefficient of correlation is independent of the origin and scale of measurement. (12)
- Q 7. (a) What are the necessary assumption you need regarding Binomial Distribution? In a small town, there are 500 families. Each family consists of four children (boy and/or girls) and their parents. For a single family find the probability that there is (10)
- (i) No boy,
- (ii) At least one boy

In that town how many families you may expect that there is no girl?

- (b) Suppose there are 11000 students in any university admission test. Let the average score is 60 (in percentage) and standard deviation is 10. Also assume that scores are normally distributed. Find the expected number of students who are (12)
- (i) Not qualified (score < 40)
- (ii) Qualified but not eligible for choosing all departments ($40 \leq \text{score} < 80$),
- (iii) Qualified and also eligible for choosing any departments (score ≥ 80)

It is noted that the required probability of each case should be shown in figure.

- (c) Fit the following data to binomial distribution. Assume that the probability of success is 0.60. (13)
- Now out of 500 students, according to the model (Binomial distribution), find the expected number of students who will obtain the score 03.

Score	0	1	2	3
No. of student	3	8	13	6

Also test its goodness of fit with 5% level of significance.

- Q 8. (a) Find the coefficient of correlation between temperature and strength of bonding of a solid material. The experimental data are given below: (15)

Temperature	10	15	20	30
Strength	3	8	13	25

- (b) Plot the following data. Find the linear regression model of score on time spent by using following observation: (20)

Time Spent	30	50	80	90
Score	3	8	11	18

Find total variation, standard error, coefficient of determination. Also estimate the score (accordingly to the model) if time spent in 200 units.

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 2nd Term (Regular) Examination, 2018
Department of Materials Science and Engineering

Ph 2227

Basic Quantum Mechanics and Solid State Physics

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Derive Einstein's photoelectric equation and explain the laws of photoelectric emission. (10)
(b) What is Compton effect? Explain and derive an expression for Compton shift on the basis of quantum theory. Discuss the results. (15)
(c) X-rays of wavelength 0.45×10^{-10} m are scattered by free electrons in a block of carbon through 90° . Find the momentum of (i) incident photons (ii) scattered photons (iii) recoil electrons and (iv) the energy of the recoil electrons. (10)
- Q 2. (a) Discuss vector atom model. Write down the names of all quantum associated with vector atom model. (12)
(b) Define wave velocity and group velocity for de Broglie waves. Show that the phase velocity is the same as group velocity. (13)
(c) Experiments indicate that 13.6 eV is required to separate a hydrogen atom into a photon and electron, that is its binding energy $E = -13.6$ eV. Find the orbital radius and velocity of the electron in a hydrogen atom. (10)
- Q 3. (a) If $\psi(x,t) = Ae^{i(kx-\omega t)}$ be a plane monochromatic wave function, then (15)
(i) Construct the time dependent Schrödinger equation
(ii) Obtain the time part solution of the equation
(b) State the postulates of Quantum Mechanics. (10)
(c) What is commutation? Using commutations relation on ψ , show that, (10)
(i) $[\hat{y}, \hat{p}_y] = i\hbar$
(ii) $[\hat{l}_x, \hat{z}] = -i\hbar \hat{y}$
- Q 4. (a) What are the physical significances of the Ladder operators $\hat{a}$ and $\hat{a}^\dagger$. For a linear harmonic oscillator, use ladder operators and show that, (15)
$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a} \hat{a}^\dagger + \hat{a}^\dagger \hat{a})$$

Where the symbols have their usual meaning.
(b) Obtain the ground state wave function ψ_0 and energy E_0 of a linear harmonic oscillator. (12)
(c) If the orbital angular momentum is expressed as $\vec{l} = \vec{r} \times \vec{p}$, then obtain all the angular momentum operators in 3D system. (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) Explain the concept of phonon. Give an evidence for the existence of phonon. (10)
- (b) Derive the vibrational modes of a diatomic linear lattice. Name the different branches of the dispersion relation curve. What is the difference between the two branches? (15)
- (c) Compare the frequencies of sound waves of wavelength, $\lambda=10^{-7}$ cm for (i) a homogeneous line (ii) acoustic waves on a linear lattice containing two identical atoms per primitive cell of interatomic spacing $2.5 \text{ \AA}$ and (iii) light waves of the same wavelength. (10)
- Given that $v_o=10^5$ cm/sec

- Q 6. (a) Show that, according to Einstein model, lattice heat capacity can be written as– (15)

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E/T}}{(e^{\theta_E/T} - 1)^2}$$

Hence, at low temperature, compare the result with experimental observations.

- (b) Obtain an expression for electrical conductivity of a metal on the basis of free electron theory. Hence prove ohm's law. (10)
- (c) Show that average kinetic energy of a free electron at 0 K is $\frac{3}{5} E_f$, where E_f is Fermi energy and average speed is $\frac{3}{4} v_f$, where v_f is the velocity at Fermi surface. (10)
- Q 7. (a) What do you understand by nanoscale, nanoscience and nanotechnology? Give at least some physical properties and chemical properties of nanomaterials. Mention risk assessment requires in nanomaterials. (13)
- (b) What is Hall effect? Derive the value of Hall voltage and Hall co-efficient R_H is given by $R_H=1/ne$ in e.m.u. Describe an experimental setup for the measurement of the Hall voltage. (12)
- (c) Find the Hall co-efficient and electron mobility for Germanium, if for a given sample (length 1 cm, breadth 3 mm and thickness 1 mm) a current of 4.2 mA is flown from a 1.5 V supply develops a Hall voltage of 20 mV across the specimen in a magnetic field of 0.45 wb/m^2 . (10)

- Q 8. (a) Discuss the followings: (12)
- (i) Population inversion and
- (ii) Stimulated emission
- (b) Explain the principle and construction of "Helium-Neon" LASER. (13)
- (c) A 1500 KW LASER emits light of 900 nm wavelength. Calculate the number of photons emitted by LASER every second. (10)